

Day 18 - Lecture 18.4: Important Plot Methods

1. Overview & Core Functions

To turn raw lines into meaningful figures, Matplotlib provides essential helper methods that label, scale, and display the plot:

Method	Purpose
<code>plt.plot(x, y)</code>	Draws line plots and markers connecting coordinates
<code>plt.title("...")</code>	Sets the main title centered above the plot
<code>plt.xlabel("...")</code>	Sets descriptive label for the horizontal axis
<code>plt.ylabel("...")</code>	Sets descriptive label for the vertical axis
<code>plt.grid(True)</code>	Displays grid lines to facilitate reading exact values
<code>plt.xlim(min, max)</code>	Sets manual boundaries for the X-axis
<code>plt.ylim(min, max)</code>	Sets manual boundaries for the Y-axis
<code>plt.show()</code>	Flushes and renders the figure to the screen / notebook

2. Practical Case Study: Oscar Winner Revenues

Tracking the box office revenue of Academy Award (Oscar) Best Picture winners:

- 2008: *The Dark Knight* (\$1005M)
- 2009: *The Hurt Locker* (\$170M)
- 2010: *The King's Speech* (\$427M)
- 2011: *The Artist* (\$133M)
- 2012: *Argo* (\$232M)

```
import matplotlib.pyplot as plt

oscar_years = [2008, 2009, 2010, 2011, 2012]
oscar_revenue = [1005, 170, 427, 133, 232] # in $M

plt.figure(figsize=(8, 5))
plt.plot(oscar_years, oscar_revenue)

# Adding labels and title
plt.title("Oscar Years vs Revenue (in $M)", fontsize=14, fontweight='bold')
plt.xlabel("Years", fontsize=12)
plt.ylabel("Revenue (in $M)", fontsize=12)
plt.grid(True, linestyle=':', alpha=0.6)
plt.show()
```

3. Key Takeaways

- Without `xlabel` and `ylabel`, a chart is uninterpretable to stakeholders. Always include units (e.g. (in \$M) or (in kg)).
- Calling `plt.show()` tells Matplotlib that figure definition is finished and ready for display.